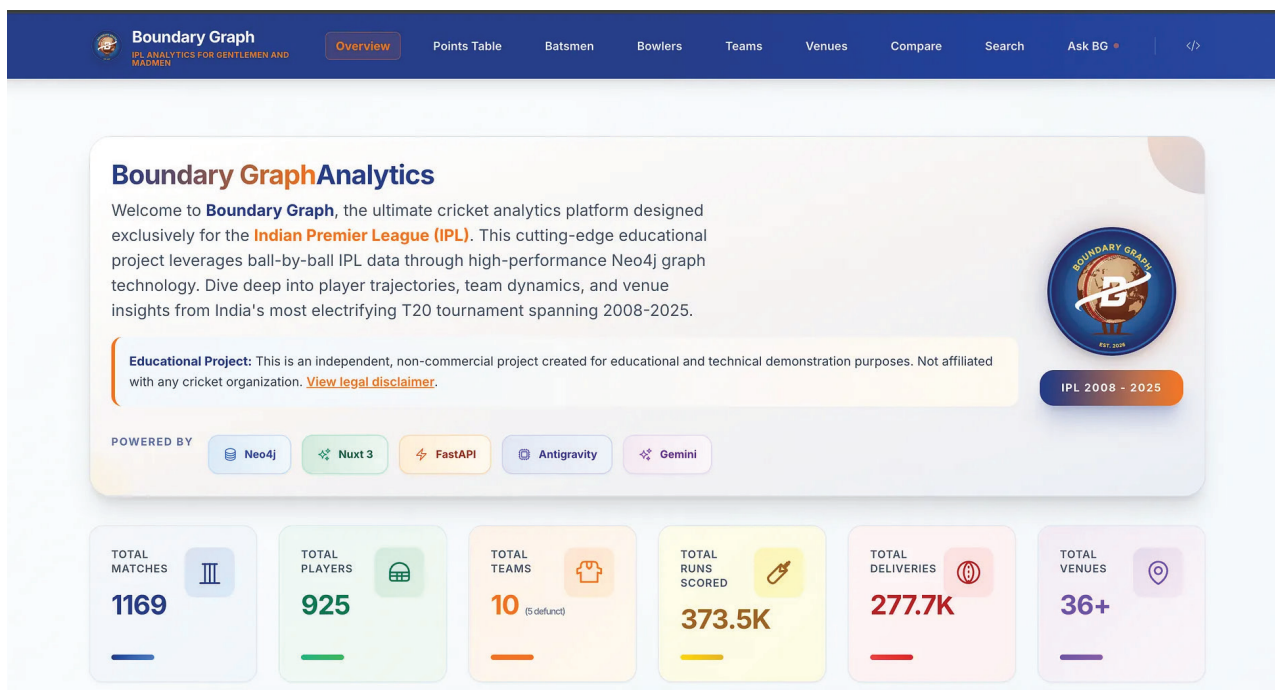


Modelling the IPL as a Graph System

Performance analyses of cricketers and teams playing in the Indian Premier League (IPL) can benefit immensely from a graph model. Here's how...



Cricket analytics in the IPL (Indian Premier League) has traditionally evolved around summarisation. Batting averages, strike rates, economy numbers, and points tables provide convenient abstractions, but they compress a highly relational sport into static outcomes. Every performance in the IPL is shaped by a web of dependencies: the opposition, the batting order, the venue, team composition, and even the strategic intent of a particular season.

What often gets lost is continuity. Players move franchises, roles change, venues behave differently across years, and teams reinvent themselves every auction cycle. These are not isolated events. They are transitions, and transitions are fundamentally relational.

A graph model embraces this reality by making relationships explicit rather than inferred.

Why graph thinking fits the IPL

The IPL is not a dataset that sits still. Squads change annually, impact players alter match dynamics, and franchises experiment with long-term versus short-term strategies. Asking meaningful questions across seasons quickly becomes cumbersome in tabular systems because relationships must be reconstructed repeatedly.

Graph thinking allows analysis to begin from connections rather than records. It becomes possible to traverse how a player's career arcs across teams, how team cores evolve, or how venue conditions interact with playing styles.

This shift enables questions such as:

- How does a player's performance change when moving between franchises with different team cores?
 - Which teams maintain structural stability across seasons despite roster churn?
 - How strongly do venues influence specific player archetypes over time?
- These are difficult to express cleanly without a relationship-native model.

Structuring the IPL as a graph

The graph is built around stable entities that rarely change identity:

- Players representing individual cricketers
- Teams representing IPL franchises
- Matches as discrete competitive events